

Tushar Kumar Kannoja

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Education

Indian Institute of Information Technology, Design and Manufacturing (IIITDM) Jabalpur 2023 – 2027
Bachelor of Technology in Computer Science and Engineering Jabalpur, India

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Computer Networks, Operating Systems, Object-Oriented Programming, Software Engineering, Foundations of Virtual and Augmented Reality

Experience

Domain Lead – Testing, Fusion (IIITDMJ College Website) Jabalpur, India

- Served as Domain Lead – Testing for Fusion, IIITDM Jabalpur's official college website, leading a cross-functional testing team responsible for QA during development.
- Drove improvements in platform stability and workflow efficiency across the website's development pipeline.

Projects

GenRAG — *AI-Powered Multi-Document Retrieval-Augmented Generation System* — Python, Streamlit, Sentence-BERT, Gemini API, Docker

- Built a Retrieval-Augmented Generation pipeline entirely from scratch, without relying on high-level frameworks like LangChain or external vector database servers.
- Implemented Sentence-BERT (all-mpnet-base-v2) semantic embeddings with recursive text splitting and chunking, indexing 1,400+ vector text chunks in a lightweight CSV-based store for sub-millisecond dot-product similarity retrieval.
- Engineered multi-PDF document ingestion with automatic indexing and direct upload support through the web interface, enabling multi-document semantic search.
- Integrated Google Gemini 1.5 with dynamic API key loading and fallback model handling; designed and deployed a glassmorphic cyberpunk-themed Streamlit UI, containerized with Docker for multi-cloud deployment.

NutriLens — *AI-Powered Nutritional Analysis Application* — React, Vite, FastAPI, TensorFlow, SQLite, Gemini API

- Engineered an EfficientNetB0-based food recognition model achieving 90.87% classification accuracy across diverse food categories.
- Built an end-to-end meal-analysis pipeline supporting image upload, portion-size adjustment, and real-time calorie and macronutrient computation.
- Integrated Gemini API for personalized nutritional recommendations and architected a full-stack React + FastAPI application for low-latency inference.

CampusOps — *Smart Campus Operations & Incident Management System* — React, Node.js, Express, PostgreSQL, Prisma, Socket.IO, Redis, BullMQ, Groq

- Engineered a real-time campus incident platform with AI-powered classification, priority/department assignment, duplicate detection, evidence management, and complete incident lifecycle tracking.
- Implemented Socket.IO WebSocket rooms with JWT authentication for live status updates, incident chat, notifications, and emergency GPS panic broadcasts with sub-200ms response targets.
- Built SLA enforcement using BullMQ + Redis delayed jobs, automatic escalation workflows, crash-safe job processing, and priority-based deadlines to improve operational accountability.
- Developed a React/Vite admin dashboard with PostgreSQL/Prisma, RBAC, Redis caching, AI summaries, hotspot detection, sentiment analysis, Jest/Supertest testing, and Cloudinary evidence storage.

Skills Summary

Languages: C++, Python, JavaScript, TypeScript, PHP, SQL, Java

Frontend: React.js, Tailwind CSS, Vite

Backend: Node.js, Express.js, FastAPI

Databases: PostgreSQL, MongoDB, MySQL, SQLite

AI/ML: TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Sentence-BERT

Tools: Git, GitHub, Docker, Redis, Prisma, Socket.IO

Achievements and Certifications

- **Uber SHE++:** Selected as one of the top 100 candidates from a competitive applicant pool of 4,000+ (top 2.5%).
- **Competitive Programming:** Solved 600+ data structures and algorithms problems across LeetCode, Codeforces, and CodeChef, strengthening core problem-solving and optimization skills.
- **NPTel Certifications:** Completed online certifications in Business Intelligence and Analytics and Foundation for Virtual and Augmented Reality system.

Leadership / Extracurricular

- **HackByte 2.0 (Organizing Committee):** Managed end-to-end logistics, scheduling, and cross-team coordination for Central India's largest hackathon, ensuring smooth execution for hundreds of participants.
- **Enthuse – College Sports Fest (Event Coordinator):** Coordinated scheduling and operations for college-wide sports events while actively competing in inter-college basketball tournaments.
- **Coordinator, Basketball Club:** Organized practice sessions and represented the club in inter-college tournaments.